Practice Exam 3 Solution

STAT 3113 Regression Analysis

The Boston Housing Dataset

In R, the MASS library contains Boston data set, which has 506 rows and 14 columns, which records medv (median house value) and 13 predictors for 506 neighborhoods around Boston. The data frame contains the following columns:

- crim —per capita crime rate by town.
- zn — proportion of residential land zoned for lots over 25,000 sq.ft.
- indus — proportion of non-retail business acres per town.
- chas — Charles River dummy variable (= 1 if tract bounds river; 0 otherwise).
- nox — nitrogen oxides concentration (parts per 10 million).
- rm — average number of rooms per dwelling.
- age — proportion of owner-occupied units built prior to 1940.
- dis — weighted mean of distances to five Boston employment centres.
- rad — index of accessibility to radial highways.
- tax — full-value property-tax rate per $10,000.
- ptratio — pupil-teacher ratio by town.
- black — $1000(Bk - 0.63)^2$ where Bk is the proportion of blacks by town.
- lstat — lower status of the population (percent).
- medv — median value of owner-occupied homes in $1000s. **(dependent variable)**

Load the Data

```
library(MASS)
names(Boston)
```

```
##  [1] "crim"    "zn"      "indus"   "chas"    "nox"     "rm"      "age"
##  [8] "dis"     "rad"     "tax"     "ptratio" "black"   "lstat"   "medv"
```

Question 1

Conduct a stepwise regression analysis of the Boston data using R to find the "best" predictors of medv. Please use 0.15 for both α-to-remove and α-to-enter. Comment on the output.

```
library(olsrr)
fit_complete<-lm(medv~., data=Boston)
ols_step_both_p(fit_complete, details=FALSE, pent=0.15, prem = 0.15)
```

```
##
##                                 Stepwise Selection Summary
```

```
## ---------------------------------------------------------------------------------------
##                         Added/                  Adj.
## Step    Variable       Removed     R-Square    R-Square      C(p)         AIC        RMSE
## ---------------------------------------------------------------------------------------
##    1      lstat        addition       0.544       0.543    362.7530    3288.9750    6.2158
##    2       rm          addition       0.639       0.637    185.6470    3173.5423    5.5403
##    3    ptratio        addition       0.679       0.677    111.6490    3116.0973    5.2294
##    4      dis          addition       0.690       0.688     91.4850    3099.3590    5.1386
##    5      nox          addition       0.708       0.705     59.7540    3071.4386    4.9939
##    6      chas         addition       0.716       0.712     47.1750    3059.9390    4.9326
##    7     black         addition       0.722       0.718     37.0590    3050.4384    4.8818
##    8       zn          addition       0.727       0.722     30.6240    3044.2750    4.8474
##    9      crim         addition       0.729       0.724     28.4180    3042.1546    4.8326
##   10      rad          addition       0.734       0.729     20.2660    3034.0687    4.7895
##   11      tax          addition       0.741       0.735     10.1150    3023.7264    4.7362
## ---------------------------------------------------------------------------------------
```

Answer:

- Response variable – medv
- Possible predictors – 13 the other variables.

From the R output, the best predictors of medv are: lstat, rm, ptratio, dis, nox, chas, black, zn, crim, rad, tax.

Question 2

What are the dangers associated with drawing inferences from the stepwise model?

Answer: The model was selected as the best model after many different models were fit to the data. There are two main dangers associate with using and interpreting this model:

1. The overall probability with making at least one Type I error is extremely high.
2. This model only includes linear components. Before using this model, both quadratic and interaction relationships in the model should be considered.

Question 3

Use all-possible-regressions-selection to find the "best" predictors of medv. Include the adjusted r-square plot and Cp plot, and illustrate how the choice is made.

```
library(leaps)
all.possible<-regsubsets(medv~., data=Boston, nvmax=13)

bestsubset.summary<-summary(all.possible)
bestsubset.summary$outmat
```

```
##           crim zn  indus chas nox rm  age dis rad tax ptratio black lstat
## 1  ( 1 )  " "  " " " "   " "  " " " " " " " " " " " " " "     " "   "*"
## 2  ( 1 )  " "  " " " "   " "  " " "*" " " " " " " " " " "     " "   "*"
## 3  ( 1 )  " "  " " " "   " "  " " "*" " " " " " " " " "*"     " "   "*"
## 4  ( 1 )  " "  " " " "   " "  " " "*" " " "*" " " " " "*"     " "   "*"
## 5  ( 1 )  " "  " " " "   " "  "*" "*" " " "*" " " " " "*"     " "   "*"
## 6  ( 1 )  " "  " " " "   "*"  "*" "*" " " "*" " " " " "*"     " "   "*"
## 7  ( 1 )  " "  " " " "   "*"  "*" "*" " " "*" " " " " "*"     "*"   "*"
## 8  ( 1 )  " "  "*" " "   "*"  "*" "*" " " "*" " " " " "*"     "*"   "*"
## 9  ( 1 )  "*"  " " " "   "*"  "*" "*" " " "*" "*" " " "*"     "*"   "*"
## 10  ( 1 ) "*"  "*" " "   " "  "*" "*" " " "*" "*" "*" "*"     "*"   "*"
## 11  ( 1 ) "*"  "*" " "   "*"  "*" "*" " " "*" "*" "*" "*"     "*"   "*"
## 12  ( 1 ) "*"  "*" "*"   "*"  "*" "*" " " "*" "*" "*" "*"     "*"   "*"
## 13  ( 1 ) "*"  "*" "*"   "*"  "*" "*" "*" "*" "*" "*" "*"     "*"   "*"
```

```
rsq = bestsubset.summary$rsq
adjrsq = bestsubset.summary$adjr2
rss = bestsubset.summary$rss
cp = bestsubset.summary$cp

p=seq(1, 13, 1)
cbind(p, adjrsq)
```

```
##        p    adjrsq
##  [1,]  1 0.5432418
##  [2,]  2 0.6371245
##  [3,]  3 0.6767036
##  [4,]  4 0.6878351
##  [5,]  5 0.7051702
##  [6,]  6 0.7123567
##  [7,]  7 0.7182560
##  [8,]  8 0.7222072
##  [9,]  9 0.7252743
## [10,] 10 0.7299149
## [11,] 11 0.7348058
## [12,] 12 0.7343282
## [13,] 13 0.7337897
```

```
cbind(p, cp)
```

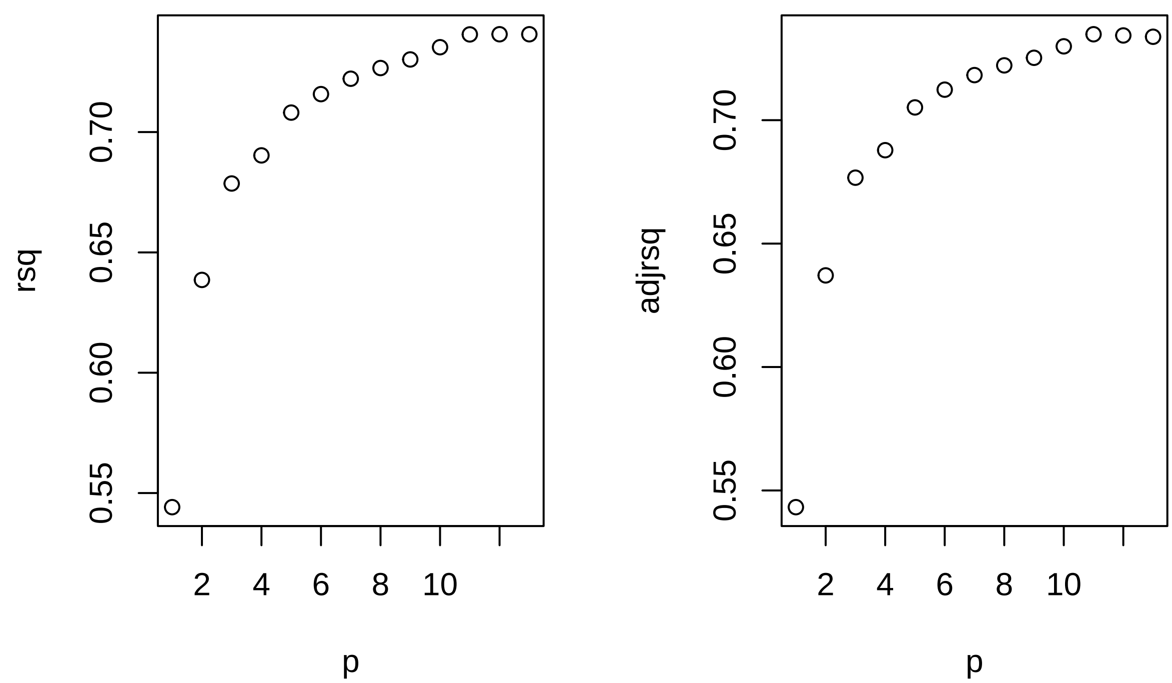
```
##        p        cp
##  [1,]  1 362.75295
##  [2,]  2 185.64743
##  [3,]  3 111.64889
##  [4,]  4  91.48526
##  [5,]  5  59.75364
##  [6,]  6  47.17537
##  [7,]  7  37.05889
##  [8,]  8  30.62398
##  [9,]  9  25.86592
## [10,] 10  18.20493
## [11,] 11  10.11455
## [12,] 12  12.00275
## [13,] 13  14.00000
```

```r
cbind(p, rss)
```

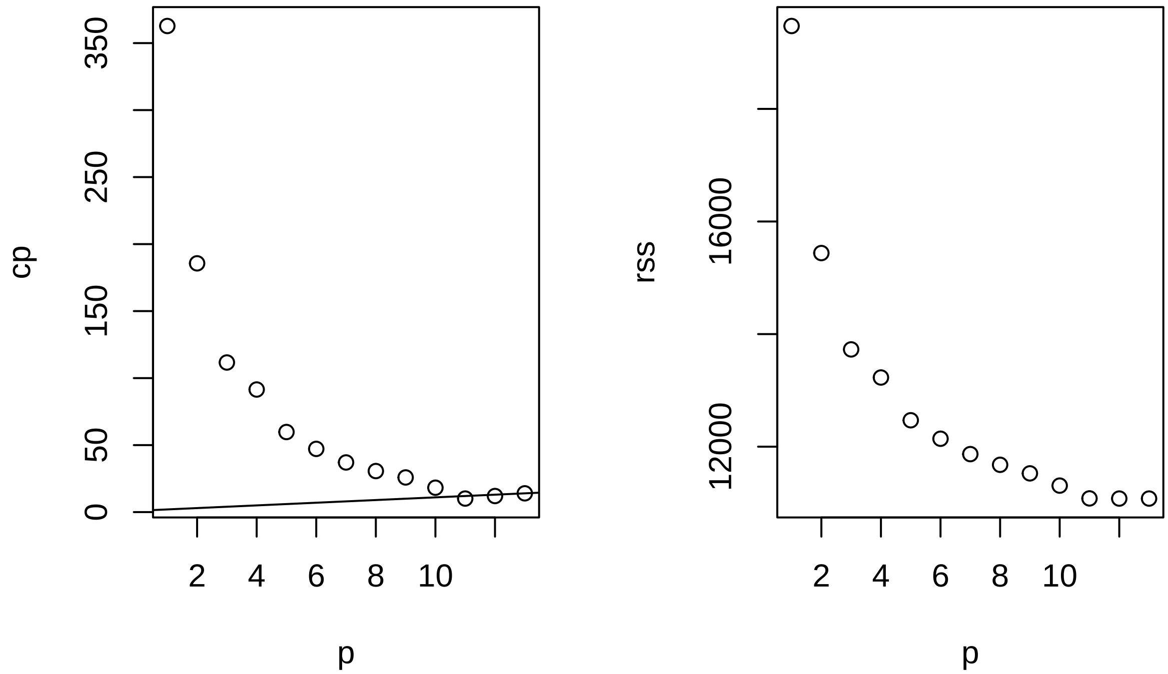
```
##        p      rss
##  [1,]  1 19472.38
##  [2,]  2 15439.31
##  [3,]  3 13727.99
##  [4,]  4 13228.91
##  [5,]  5 12469.34
##  [6,]  6 12141.07
##  [7,]  7 11868.24
##  [8,]  8 11678.30
##  [9,]  9 11526.12
## [10,] 10 11308.58
## [11,] 11 11081.36
## [12,] 12 11078.85
## [13,] 13 11078.78
```

```r
### Plot the rsq, adjrsq, rss, cp vs p

par(mfrow=c(1,2))
plot(p, rsq)
plot(p, adjrsq)
```



```r
plot(p, cp)
abline(1,1)
plot(p, rss)
```




```r
par(mfrow=c(1,1))
```

Answer: The larger the value of R_a^2, the better. On the other hand, we prefer models with a small value of C_p and a value of C_p near $p+1$. Based on these criterion, we prefer the model with 11 predictors.

The 11 predictors are: lstat, rm, ptratio, dis, nox, chas, black, zn, crim, rad, tax.

Question 4

Compare the results in Question 1 and 3, which independent variables consistently are selected as the "best" predictors?

Answer: Based on the Question 1 and 3, the 11 independent variables are consistently chosen as the best predictors are: lstat, rm, ptratio, dis, nox, chas, black, zn, crim, rad, tax.

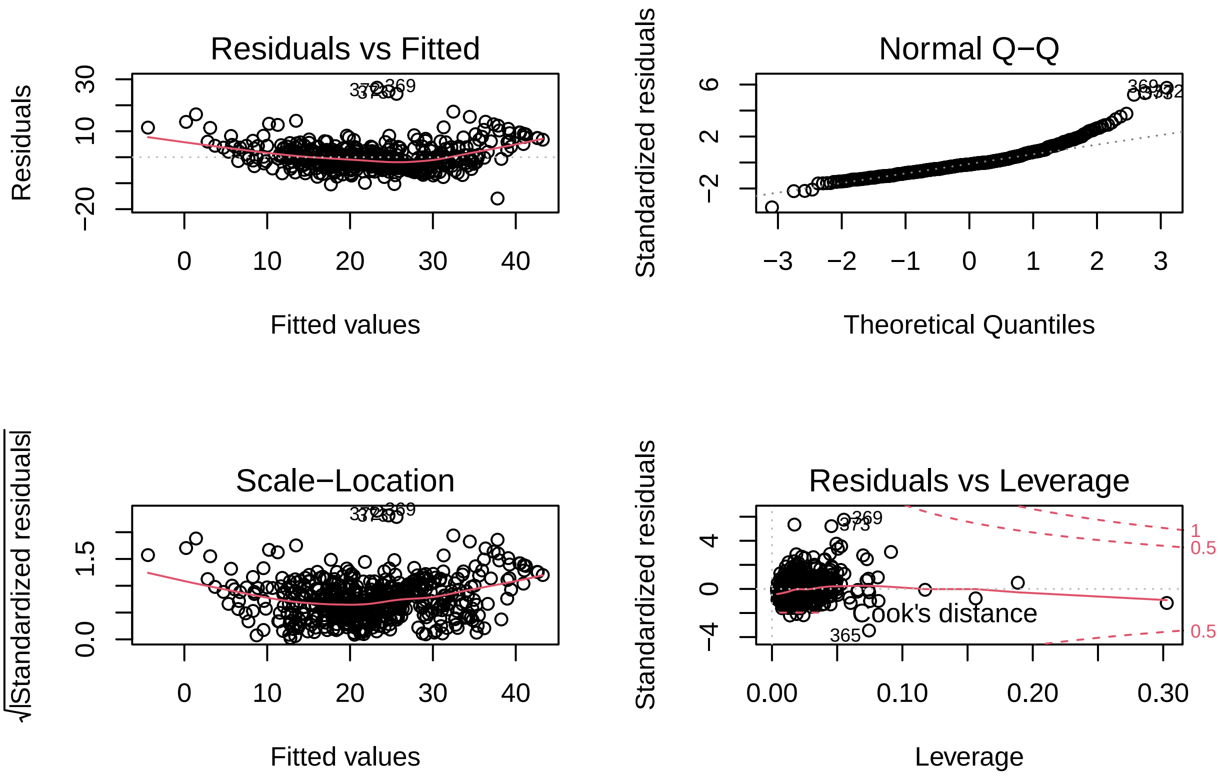
Question 5

Fit the first-order linear model with 10 independent variables: crim, chas, nox, rm, dis, rad, tax, ptratio, black, lstat. Plot the residual plots. Comment on the four residual plots one by one on the issues you found.

What model adjustments would you recommend?

```
fit = lm(medv~crim + chas + nox + rm + dis + rad + tax + ptratio + black +  lstat, data = Boston)

par(mfrow=c(2,2))
plot(fit)
```



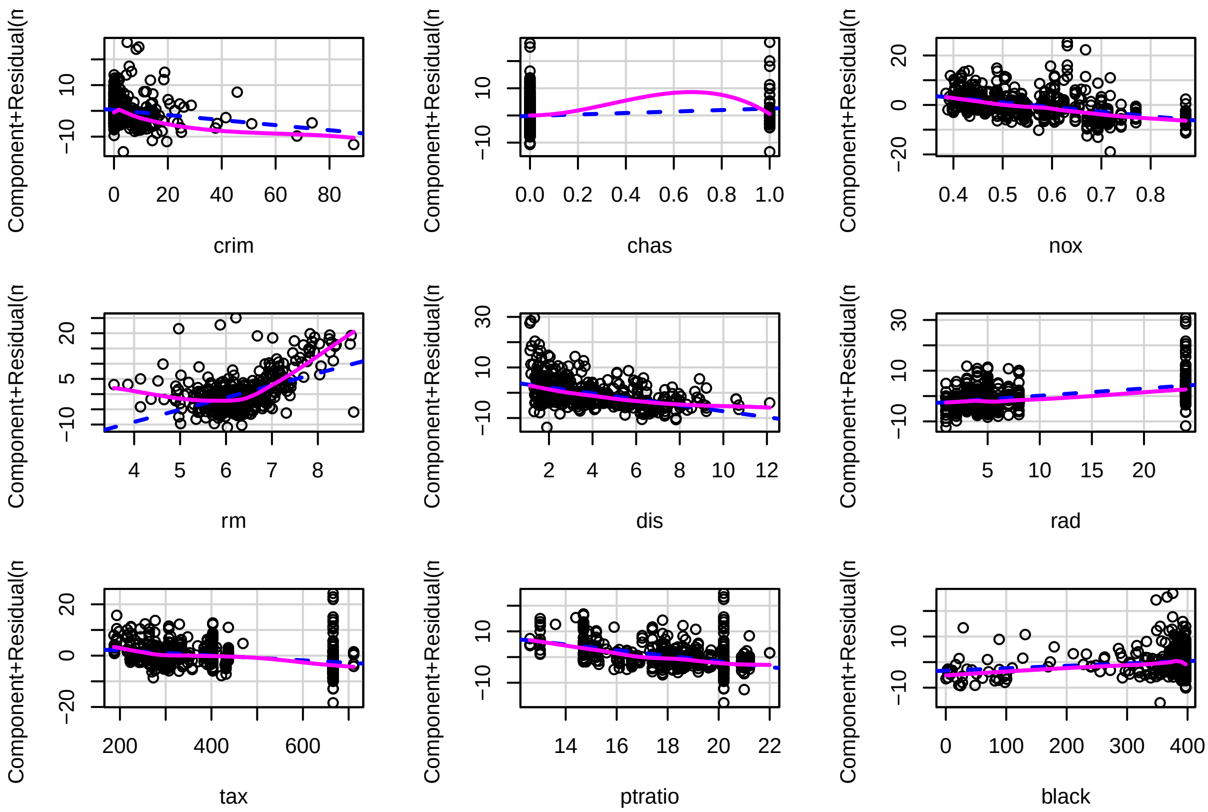

```
par(mfrow=c(1,1))
```

Answer: From the residual plot in the upper left corner, we see a U-shape or parabolic trend, which implies the model has the lack of fit issue. We might consider adding the second order terms of predictors or use partial residual plots to get more information on the relationship between y and the predictors.

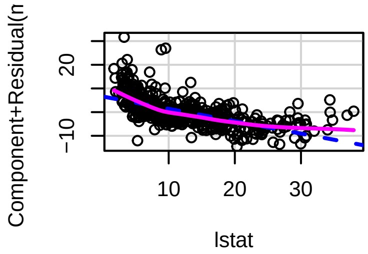
Question 6

Use the following code to plot the partial residual plots. Comment on the partial residual plots. What do the plots reveal the information between medv and the independent variables?

```
library(car)
crPlots(fit)
```


Component+Residual(n
crim
chas
nox
rm
dis
rad
tax
ptratio
black

Component + Residual Plots



Answer: From the partial residual plots above, we could find the medv vs rm, and medv vs lstat has significant curvature.

We should consider adding the quadratic terms of rm and lstat.

Question 7

Fit the model with 10 variables in Question 5 and add second-order term of lstat, second-order term of rm, and the interaction between rad & lstat, rm &rad, crim & chas, chas & nox.

```
fit.c = lm(medv~crim + chas + nox + rm + dis + rad + tax + ptratio + black +  lstat
           + I(lstat^2) + I(rm^2) + rad:lstat + rm:rad + crim:chas + chas:nox,
           data = Boston)
summary(fit.c)
```

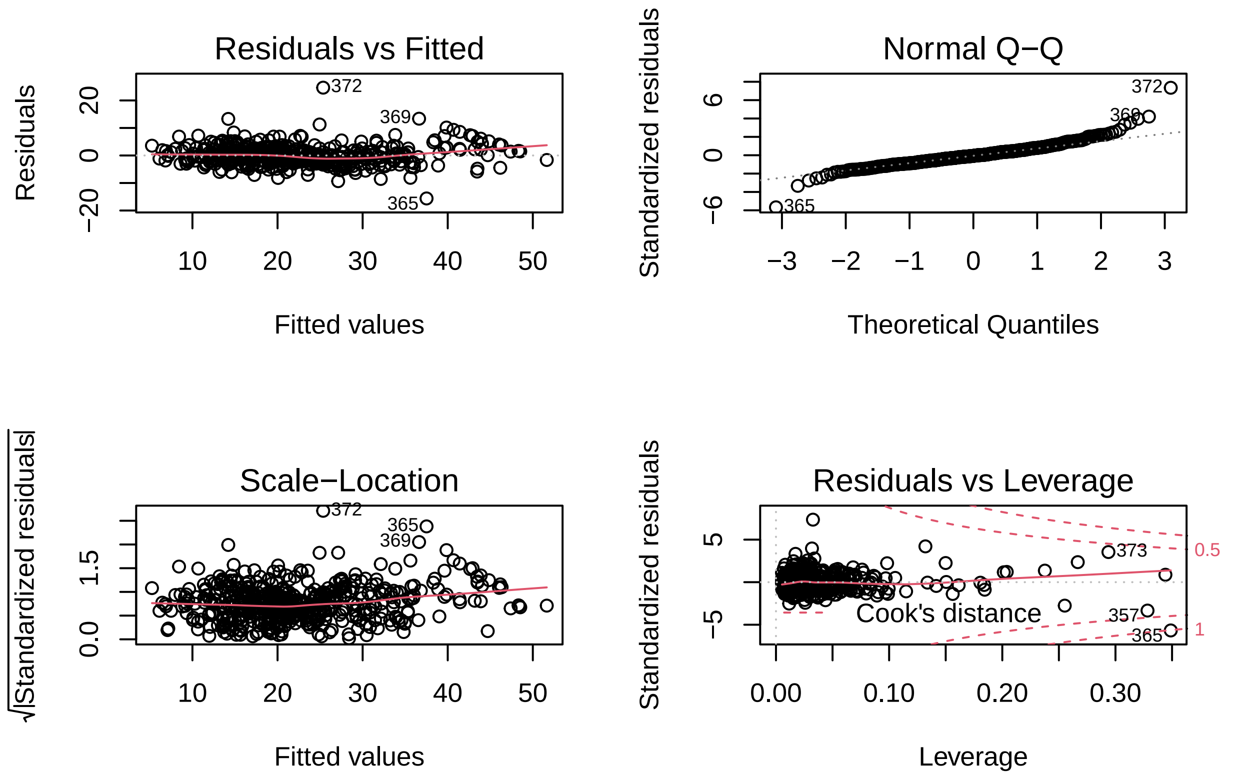
```
##
## Call:
## lm(formula = medv ~ crim + chas + nox + rm + dis + rad + tax +
##     ptratio + black + lstat + I(lstat^2) + I(rm^2) + rad:lstat +
##     rm:rad + crim:chas + chas:nox, data = Boston)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -15.610  -2.026  -0.223   1.604  24.638
```

```
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  42.046646  11.485042   3.661 0.000279 ***
## crim         -0.111171   0.024327  -4.570 6.19e-06 ***
## chas         13.473038   3.087684   4.363 1.56e-05 ***
## nox         -12.363598   2.764480  -4.472 9.63e-06 ***
## rm           -4.427683   3.056792  -1.448 0.148126
## dis          -0.820218   0.120105  -6.829 2.54e-11 ***
## rad           2.802634   0.264222  10.607  < 2e-16 ***
## tax          -0.008726   0.002397  -3.640 0.000301 ***
## ptratio      -0.745128   0.090980  -8.190 2.29e-15 ***
## black         0.003524   0.001948   1.809 0.071130 .
## lstat        -0.881908   0.109460  -8.057 6.04e-15 ***
## I(lstat^2)    0.024918   0.003128   7.967 1.15e-14 ***
## I(rm^2)       0.973805   0.216680   4.494 8.73e-06 ***
## rad:lstat    -0.035154   0.003714  -9.466  < 2e-16 ***
## rm:rad       -0.338581   0.037780  -8.962  < 2e-16 ***
## crim:chas     2.141012   0.322928   6.630 8.90e-11 ***
## chas:nox    -26.028896   5.684180  -4.579 5.93e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.41 on 489 degrees of freedom
## Multiple R-squared:  0.8669, Adjusted R-squared:  0.8625
## F-statistic:   199 on 16 and 489 DF,  p-value: < 2.2e-16
```

```
source("anova_alt.R")
anova_alt(fit.c)
```

```
## Analysis of Variance Table
##
##         Df    SS      MS      F           P
## Source  16 37030 2314.36 199.01 3.2656e-202
## Error  489  5687   11.63
## Total  505 42716   84.59
```

```
par(mfrow=c(2,2))
plot(fit.c)
```




```r
par(mfrow=c(1,1))
```

Comment on the model fitted.

Answer: From the output above, we could see that overall F-test yield significant conclusion. What's more, the $R_a^2 = 86.25\%$ and $SE = 3.41$. The overall model is statistically useful for predicting medv.

We then examine the residual plots.

- From the residual plot, there is no significant patterns residual plots in the upper left corner.
- The Normal QQ plot generally looks OK.
- The plot in the lower left corner doesn't show significant increasing or decreasing trend, which suggests homogeneous variance. By using standardized residual method, there are outliers.
- Observation 365 is influential.

Question 8

For the model fitted in Question 7, is the normality assumption reasonably satisfied?

Answer: Yes, based on the Normal QQ plot in Question 7, the normality assumption is reasonably satisfied.

Question 9

Use Studentized Deleted Residual method to identify whether there are any outliers.

```
studresid = rstudent(fit.c)
studresid_sorted = sort(studresid)
head(round(studresid_sorted, 3),10)
```

```
##    365    357    366     89    506    427    343    309     11    398
## -5.863 -3.372 -2.775 -2.530 -2.437 -2.129 -2.104 -1.887 -1.814 -1.809
```

```
tail(round(studresid_sorted,3),10)
```

```
##   370   153   392   187   162   182   373   410   369   372
## 2.261 2.365 2.481 2.567 2.802 3.365 3.583 4.022 4.281 7.780
```

Answer: From the R output above, we could find there are several outliers: Observation #365, #357, #182, #373, #410, # 369, # 372, which all have the studentized deleted residuals with absolute value greater than 3.

Our model doesn't contain all the significant interactions. This might be the reason that there are several outliers.

Question 10

Use Cook's Distance method to identify whether there are any influential observations.

```
cookdist = cooks.distance(fit.c)
cooks_sorted = sort(round(cookdist,3))

tail(round(cooks_sorted,3),10)
```

```
##   413   155   370   372   153   366   369   373   357   365
## 0.032 0.034 0.053 0.107 0.119 0.153 0.158 0.307 0.320 1.014
```

Answer: The model fitted in Question 7 has 16 independent variables, thus $k = 16$. There are 506 observations. Thus, $n = 506$.

Thus $\nu_1 = 16 + 1$, $\nu_2 = n - (k + 1) = 506 - (16 + 1) = 489$. Therefore,

$$F_{0.5,17,489} = 0.96$$

Therefore, observation 365 has cook's distance $= 1.014 > 0.96$. We have observation 365 is an influential observation. This conclusion is consistent with the conclusion in Question 7.